

Zhengyang Qi



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PROFESSIONAL SUMMARY

Researcher focused on post-training and evaluation of frontier LLMs. Experienced in reinforcement learning, scalable evaluation environments, and large-scale RLHF pipelines. Interested in building agentic RL training and evaluation.

EDUCATION

- **Carnegie Mellon University, School of Computer Science** Pittsburgh, PA
M.S. in Intelligent Information Systems (3.84/4.30) Aug 2022 - May 2024
- **Georgia Institute of Technology, Emory University, Dual Degree Program** Atlanta, GA
B.S in Computer Science, B.A. in Math and Economics (3.92/4.00) Aug 2017 - May 2022

RELEVANT PUBLICATIONS

* denotes equal contribution

- **RIFT: A Rubric Failure Mode Taxonomy and Automated Diagnostics:** (Zhengyang Qi, Charles Dickens, Derek Pham, Amanda Dsouza, Armin Parchami, Fred Sala, Paroma Varma) *ICLR 2026 Data Problems for Foundation Models Workshop*.
- **BeTaL: Adaptive Benchmarking with LLM-Guided Environment Design:** (Amanda Dsouza, Harit Vishwakarma, Zhengyang Qi, Justin Bauer, Derek Pham, Tom Walshe, Armin Parchami, Fred Sala, Paroma Varma) *Arxiv preprint*.
- **Sotopia-RL: Reward Design For Social Intelligence:** (Zhengyang Qi*, Haofei Yu*, Yining Zhao*, Keyang Xuan, Kolby Nottingham, Bodhisattwa Prasad Majumder, Hao Zhu, Paul Pu Liang, Jiaxuan You) *Arxiv preprint*.
- **MMoE: Enhancing Multimodal Models with Mixtures of Multimodal Interaction Experts:** (Haofei Yu*, Zhengyang Qi*, Lawrence Jang*, Ruslan Salakhutdinov, Louis-Philippe Morency, Paul Pu Liang) *The 2024 Conference of EMNLP. Main Conference*.
- **Sotopia- π : Interactive Learning of Socially Intelligent Language Agents:** (Ruiyi Wang*, Haofei Yu*, Wenxin Zhang*, Zhengyang Qi*, Maarten Sap, Graham Neubig, Yonatan Bisk, Hao Zhu) *The ACL 2024 Conference. Main Conference*.
- **Sotopia: Interactive Evaluation for Social Intelligence in Language Agents:** (Xuhui Zhu*, Hao Zhu*, Leena Marthur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, Maarten Sap) *The ICLR 2024 Conference. Spotlight*.
- **Long-Horizon Dialogue State Tracking for Role Identification in the Game of Avalon with Large Language Models:** Simon Stepputtis, Joseph Campbell, Yaqi Xie, Zhengyang Qi, Wenxin Zhang, Ruiyi Wang, Sanketh Rangrejji, Charles Michael Lewis, Katia P. Sycara *Findings of the EMNLP 2023 Conference*.

WORK EXPERIENCES

- **Snorkel AI** San Francisco, CA
Senior Research Scientist Sep 2025 - Present
 - **Frontier Agentic Benchmarking & Reasoning Evaluation:** Curated multiturn, human preference and rubrics pipelines for frontier coding agents. Designed and crowdsourced interactive tasks for computer-use agents to evaluate visual reasoning and grounding in specialized **EDA (Electronic Design Automation)** workflows. Developed human evaluation pipelines for clinical AI, utilizing **rubric-based preference** and **long context synthetic EHR simulations**.
 - **RL Post-Training for Agentic LLMs:** Led agentic RL training for terminal-based coding agents using **GRPO**, achieving a **10% gain(2.2x)** in performance on **Terminal Bench 2.0** (5.6% $\rightarrow$ 13.4% resolution rate). Experimented with environment **data selection**, **llm-judge**, **rubrics**, and **process reward** signals for RL training. Scaled synthetic and distillation pipelines via **rejection sampling** to optimize open source 30B models for multi-step environment grounding.
 - **Agentic RL Infrastructure:** Engineered a multi-node RL training backbone from the ground up using **verl**, scaling to **64 H200 GPUs** and managing **6,000+** rollout environments. Optimized orchestration and I/O for **10TB+** of containerized rollouts with sidecar DinD, container registry cache, overlay FS, and optimized **Ray** configuration. Accelerated **fully async RL** for long tail latency (40 mins $\rightarrow$ 7 mins per round). Implemented custom logging for **full failure mode profiling**.
- **Scale AI** San Francisco, CA
Machine Learning Research Engineer Aug 2024 - Aug 2025
 - **Post-Training of Foundational Reward Models:** Led post-training for **multi-modal LLM reward models** using **SFT** and **GRPO** to align human reward and preference. Curated **60k+** evaluation and **300k+** training data for comprehensive capabilities (multilingual, IF, multimodal reasoning, STEM, and coding). Experimented with open models including Llama 3.3 70B, Qwen2.5VL, and QwQ 32B, optimizing training recipes via data augmentation and reasoning RL to overcome spurious reward hacking. Delivered a SOTA model **outperforming GPT-4o** and deployed in production.
 - **Scalable Oversight:** Led scalable oversight initiatives for human-AI hybrid data pipelines. Deployed models as auto rating service. Designed metrics through **A/B test** for effective human labor reduction **without biasing human judgements**.
 - **Reasoning Rubrics Quality:** Developed automated quality-control agents for RLHF rubric pipelines using **RAG**, tool use, and multi-agent debate. Achieved improvements over baseline heuristics across production deployments.

TECHNICAL SKILLS

- **Languages:** Python, Unix, SQL, JavaScript
- **ML & Systems:** PyTorch, Ray, vLLM, Sglang, CUDA, Distributed Training (FSDP, Megatron, DeepSpeed), Spark
- **Frontier AI:** RLHF/RLAIF (PPO, GRPO, DPO, ReST), Reward Modeling, Rubric Evaluation, Scalable Oversight, Synthetic Data Generation, Test-Time Compute, Chain-of-Thought, Multimodal Models